AquaCanvas: Real-Time Watercolor Simulation

CS 184 Final Project — Spring 2026

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Project Goal

Build a digital painting tool that simulates watercolor behavior in real time.

Key Ideas

Water flow, pigment transport, drying, and GPU acceleration in Godot Engine 4.4.1

Overall Architecture: CPU vs GPU

CPU

User Interface input detection
brush type, color picker, water amount slider

Camera / viewport controls

Canvas drawing
detect brush position in image space

Stroke stored locally in a bitmask
temporary pixel buffer per stroke

Batched upload
accumulate dabs, then push to GPU memory in one batch call

GPU

1. Receive stroke data
update texture maps: water layer, mobile pigment layer

2. Compute shader: Evaporation
water dries over time

3. Compute shader: Force Calculation
gravity + surface tension + spreading force ⟶ 4-directional displacement map

4. Compute shader: Water & Pigment Transport
move water and carry pigment proportionally

5. Compute shader: Deposition
wet pigment settles onto static (dried) layer

6. Compute shader: Diffusion
static pigment spreads to soften edges

7. Output: updated texture
rendered to screen for next frame

Technical Challenges

1. Checkerboard Artifact

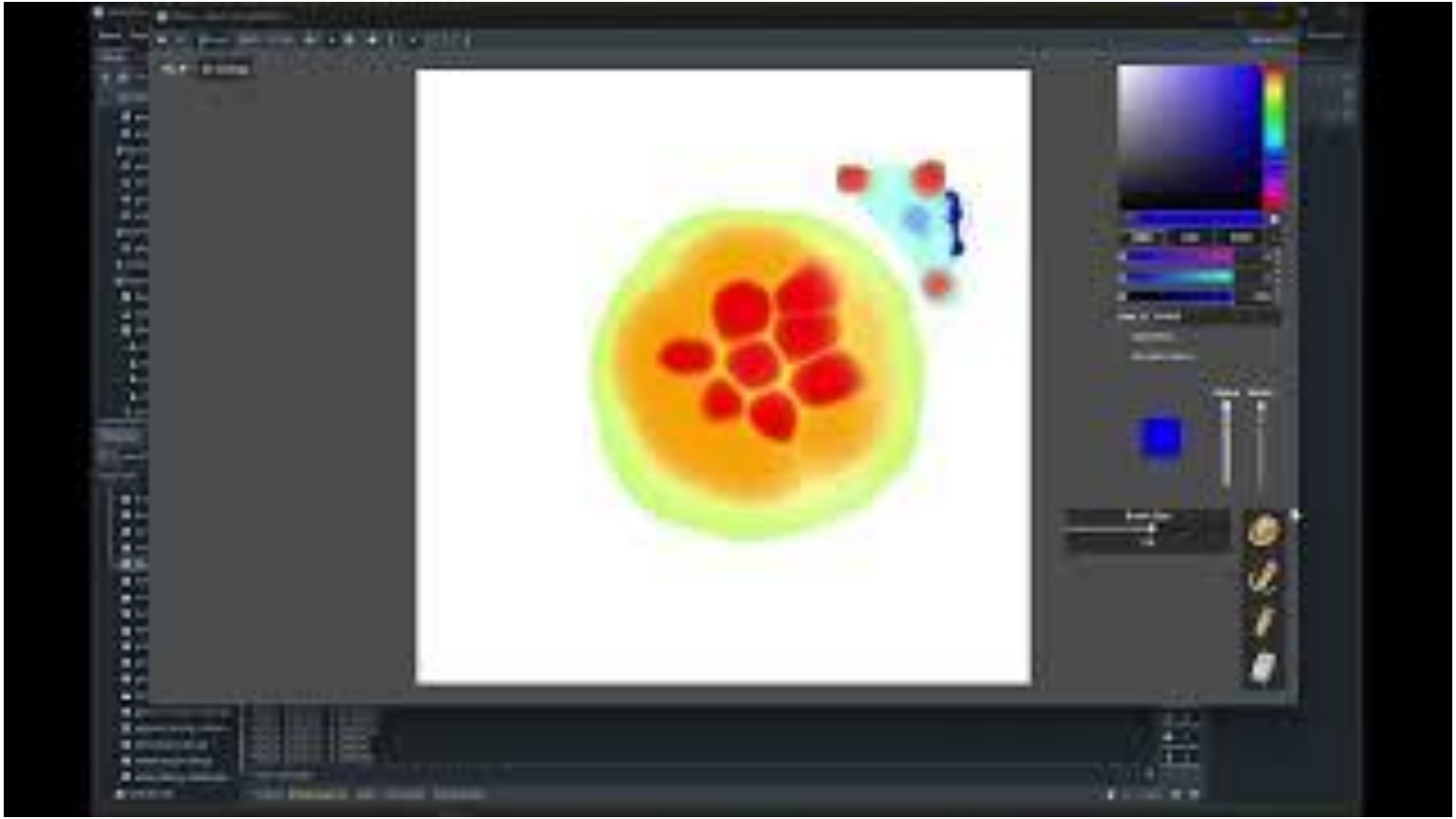
Naive water redistribution (Δx / Δy per cell) caused a flickering checkerboard pattern at our grid scale.
Fixed by tracking 4 directional forces per pixel independently in a separate displacement map, computed before any values are updated.

2. Outflow → Inflow Reformulation

Original outflow update let each pixel push water/pigment to neighbors, which caused write conflicts on the GPU.
We rewrote it as an inflow update, where each pixel gathers from neighbors instead, making the simulation parallelizable and GPU-friendly.

3. CPU→GPU Stroke Latency

Sending every brush position to the GPU while painting created heavy overhead and broke real-time interaction.
Fixed with batched uploads: brush dabs are stored locally in a pixel buffer, then pushed to GPU memory at intervals.

Sample Images

Final Video